

Anoop Hattekar

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Location: Pune, Maharashtra, India

Summary

Embedded systems enthusiast with strong foundation in **firmware development** using **C/C++** on ESP32, STM32, Arduino, PIC18F, and 8051 platforms. Experienced in peripheral interfacing, real-time control, and hardware-level debugging. Passionate about hardware-software integration with **strong electronics fundamentals** and genuine curiosity for solving embedded challenges.

Experience

Full Stack Developer Intern

February 2026 – Present

TechB Softwares

Pune, India

- Worked on backend logic and data processing in **Python/Django** for energy-based applications involving real-time data handling
- Designed and integrated **REST APIs** ensuring reliable communication between frontend and backend systems; used **Git** for version control and code reviews
- Gained exposure to scalable system design, debugging, and handling production-level codebases

Technical Skills

Programming: C, C++, Embedded C, Data Structures

MCUs (8/16/32-bit): ESP32 (32-bit), STM32 (32-bit), Arduino, 8051 (8-bit), PIC18F

Protocols: UART, USART, SPI, I2C, Bluetooth

Peripherals: ADC, PWM, Timers, GPIO

Power Supply: BUCK, BOOST converter topologies (awareness)

Test Equipment: MSO, DSO, Oscilloscope, Logic Analyzer, Multimeter, Serial Debugging

Tools: Keil uVision, STM32CubeIDE, Arduino IDE, Git, GitHub

OS: Linux (Ubuntu)

Projects

Line Follower Pick & Drop Rescue Bot | *ESP32, Raspberry Pi, Embedded C, PID, UART, USART, I2C*

2026

- Designed and developed an autonomous rescue bot using **ESP32** as the main controller and **Raspberry Pi** for real-time camera processing; implemented **PID-based control algorithm** for precise line following and dynamic path correction
- Implemented **Bluetooth** wireless communication between ESP32 and control interface; integrated multiple peripherals including **PWM motor control**, ADC, GPIO, and timers
- Applied **UART, USART, and I2C** protocols for inter-module data exchange; performed hardware-firmware validation via logic analyzer and oscilloscope-based signal verification

Smart Wheelchair System | *Arduino Nano, Embedded C/C++, I2C, PWM, Interrupts*

2024

- Developed **interrupt-driven firmware** in Embedded C/C++ for motor control using **PWM** and real-time input handling; implemented built-in diagnostics for real-time fault detection
- Interfaced motor drivers and sensors using **I2C communication**; performed serial debugging and oscilloscope-based signal verification; assembled and tested circuit boards
- Recognized with **Government of India Copyright** (Reg. No: LD-20250167659) for innovation in embedded assistive technology

Achievements

- **Project Competition Winner (2024):** Selected for SPPU State-level competition among 100+ participants
- **Government Copyright (2025):** Recognized for innovation in embedded assistive technology

Education

Bachelor of Engineering — Electronics and Telecommunication

Expected June 2027

PCET's Pimpri Chinchwad College of Engineering and Research

CGPA: 9.1/10.0

Higher Secondary Certificate (HSC)

February 2023

JK Junior College of Science

76.67%